# Lesson 1: Real Numbers, Sequences, and Series

AI/ML Mathematics

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# 1 Motivation and Overview

Consider a Markov decision process with state space $S$ and discount factor $\gamma \in (0,1)$. The Bellman optimality operator $T$ maps any bounded value function $V$ to the function

$$(TV)(s) = \max_a \Big[ r(s,a) + \gamma \sum_{s'} p(s' \mid s,a)\, V(s') \Big].$$

Value iteration generates the sequence $V_0, V_1 = TV_0, V_2 = TV_1, \ldots$ and one asserts that $V_n$ converges to the unique optimal value function $V^*$. But *why* does this limit exist? The proof relies on the Banach fixed-point theorem, which itself rests on the completeness of $\mathbb{R}$: every Cauchy sequence of real numbers converges. Without the completeness axiom, the iterates could approach a limit that simply does not exist in our number system—exactly the situation one encounters in $\mathbb{Q}$. The tools developed in this lesson will let us formalize convergence, suprema, and the completeness property that underwrites every convergence argument in reinforcement learning and optimization.

**Why RL/ML needs the real number system**

- **Value iteration.** Convergence of the sequence $V_n = T^n V_0$ to $V^*$ requires $\mathbb{R}$ to be complete (every Cauchy sequence converges).
- **Gradient descent.** The statement "$f(x_k)$ converges to $\inf_x f(x)$" relies on the infimum existing, guaranteed by the completeness axiom.
- **Monte Carlo estimation.** The law of large numbers asserts $\frac{1}{n}\sum_{i=1}^n X_i \to \mathbb{E}[X]$; its proof uses the Bolzano–Weierstrass theorem and Cauchy completeness developed here.
- **Series in function approximation.** Neural network universality theorems express target functions as convergent series; the ratio and root tests verify absolute convergence of these expansions.

**Prerequisites.** Familiarity with high-school algebra (field operations, inequalities, absolute values), basic set theory (subsets, unions, intersections), and informal notions of natural numbers $\mathbb{N}$, integers $\mathbb{Z}$, and rationals $\mathbb{Q}$. No prior analysis course is assumed.

**Learning objectives.** After completing this lesson, you will be able to:

- **State** the ordered-field axioms and the completeness axiom for $\mathbb{R}$, and **explain** why $\mathbb{Q}$ fails to be complete.
- **Define** supremum, infimum, convergence of sequences, and Cauchy sequences.
- **Prove** the Archimedean property, the monotone convergence theorem, and the Bolzano–Weierstrass theorem.
- **Compute** limits using the algebraic limit theorems and determine convergence of series via the ratio and root tests.
- **Derive** the relationship between $\limsup$, $\liminf$, and convergence of a bounded sequence.

# 2 References


- Rudin [1976, Chapters 1–3] — The canonical reference for the real number system, sequences, and series; concise and rigorous.
- Abbott [2015, Chapters 1–2] — An accessible introduction emphasizing the completeness axiom and its consequences.

- Folland [1999, Chapter 0] — A brief review of real-number foundations aimed at students proceeding to measure theory.
- Bartle and Sherbert [2011, Chapters 2–3] — Detailed treatment of sequences, series, and convergence with many worked examples.
- Tao [2016, Chapter 1] — A construction-oriented approach building $\mathbb{R}$ from the rationals via Cauchy sequences.

## 3 The Real Number System

**Definition 3.1** (Ordered Field)**.** The real numbers $\mathbb{R}$ form an *ordered field*: a set equipped with operations $+$ and $\cdot$ and a total order $\leq$ satisfying:

(i) **Field axioms.** $(\mathbb{R}, +, \cdot)$ is a field: addition and multiplication are commutative, associative, and distributive, with identity elements 0 and 1 respectively, additive inverses for all elements, and multiplicative inverses for all nonzero elements.

(ii) **Order axioms.** The relation $\leq$ is a total order compatible with the field operations: if $a \leq b$ then $a + c \leq b + c$ for all $c$; if $0 \leq a$ and $0 \leq b$ then $0 \leq ab$.

*Remark* 3.2. The rationals $\mathbb{Q}$ also form an ordered field. What distinguishes $\mathbb{R}$ from $\mathbb{Q}$ is the completeness axiom below.

**Example 3.3.** The set $S = \{q \in \mathbb{Q} : q^2 < 2\}$ is nonempty (e.g., $1 \in S$) and bounded above in $\mathbb{Q}$ (e.g., by 2). However, $S$ has no supremum *in* $\mathbb{Q}$: any candidate rational upper bound can be improved by moving closer to $\sqrt{2}$, which is irrational. This failure motivates the completeness axiom.

**Definition 3.4** (Upper Bound, Supremum)**.** Let $S \subseteq \mathbb{R}$ be nonempty.

(i) An element $u \in \mathbb{R}$ is an *upper bound* for $S$ if $s \leq u$ for every $s \in S$. We say $S$ is *bounded above*.

(ii) An element $\alpha \in \mathbb{R}$ is the *supremum* (or *least upper bound*) of $S$, written $\alpha = \sup S$, if:
   (a) $\alpha$ is an upper bound for $S$, and
   (b) if $u$ is any upper bound for $S$, then $\alpha \leq u$.

**Definition 3.5** (Lower Bound, Infimum)**.** Dually, $\ell \in \mathbb{R}$ is a *lower bound* for $S$ if $\ell \leq s$ for every $s \in S$. The *infimum* $\inf S$ is the greatest lower bound: $\inf S$ is a lower bound and $\ell \leq \inf S$ for every lower bound $\ell$.

*Remark* 3.6. The supremum $\alpha = \sup S$ is characterized by the *$\varepsilon$-criterion*: $\alpha$ is an upper bound for $S$, and for every $\varepsilon > 0$ there exists $s \in S$ with $s > \alpha - \varepsilon$. This characterization is used constantly in analysis.

**Completeness Axiom**

**Theorem 3.7** (Completeness Axiom / Least Upper Bound Property)**.** *Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.*

This is an *axiom*: it is not proved from more basic principles but is the defining property that distinguishes $\mathbb{R}$ from $\mathbb{Q}$. It guarantees that limits, infima, and suprema exist whenever they should.

**Corollary 3.8** (Greatest Lower Bound Property)**.** *Every nonempty subset of $\mathbb{R}$ that is bounded below has an infimum in $\mathbb{R}$.*

*Proof.* Let $S \subseteq \mathbb{R}$ be nonempty and bounded below. Define $-S = \{-s : s \in S\}$. Then $-S$ is nonempty and bounded above (if $\ell$ is a lower bound for $S$, then $-\ell$ is an upper bound for $-S$). By the completeness axiom, $\alpha = \sup(-S)$ exists. We claim $\inf S = -\alpha$.

Since $\alpha$ is an upper bound for $-S$, we have $-s \leq \alpha$ for all $s \in S$, hence $s \geq -\alpha$, so $-\alpha$ is a lower bound for $S$. If $\ell$ is any lower bound for $S$, then $-\ell$ is an upper bound for $-S$, so $\alpha \leq -\ell$, hence $\ell \leq -\alpha$. Thus $-\alpha$ is the greatest lower bound. □

**Suprema and infima in optimization**

In optimization and RL, one routinely writes $\inf_{x \in X} f(x)$ or $\sup_\pi V^\pi(s)$. The completeness axiom and the greatest lower bound property guarantee these quantities exist as real numbers (when the sets are bounded), rather than as formal symbols with no concrete value. Without completeness, the optimal value function $V^*(s) = \sup_\pi V^\pi(s)$ might fail to exist, and no algorithm could converge to it.

**Theorem 3.9** (Archimedean Property)**.** *For every $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ with $n > x$. Equivalently, for every $\varepsilon > 0$ and every $M > 0$, there exists $n \in \mathbb{N}$ with $n\varepsilon > M$.*

*Proof.* Suppose for contradiction that $\mathbb{N}$ is bounded above in $\mathbb{R}$. By the completeness axiom, $\alpha = \sup \mathbb{N}$ exists. Then $\alpha - 1$ is not an upper bound for $\mathbb{N}$ (since $\alpha$ is the *least* upper bound), so there exists $m \in \mathbb{N}$ with $m > \alpha - 1$. But then $m + 1 > \alpha$ with $m + 1 \in \mathbb{N}$, contradicting the fact that $\alpha$ is an upper bound for $\mathbb{N}$.

The second formulation follows by applying the first to $x = M/\varepsilon$. □

**Example 3.10.** In gradient descent with step size $\eta > 0$, one often needs $n$ iterations so that $n\eta$ exceeds a given threshold $M$ (e.g., to traverse a distance $M$ in parameter space). The Archimedean property guarantees such $n$ exists for any positive step size, no matter how small.

**Theorem 3.11** (Density of Rationals)**.** *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $q \in \mathbb{Q}$ with $a < q < b$.*

*Proof.* By the Archimedean property, choose $n \in \mathbb{N}$ with $n(b - a) > 1$, i.e., $nb - na > 1$. Again by the Archimedean property, the sets $\{k \in \mathbb{Z} : k > na\}$ and $\{k \in \mathbb{Z} : k < na\}$ are both nonempty, so we may define $m = \min\{k \in \mathbb{Z} : k > na\}$ (this minimum exists because the positive integers exceeding $na$ form a nonempty subset of $\mathbb{N}$, which has a least element by the well-ordering principle). Then $m > na$ and $m - 1 \leq na$, so $m \leq na + 1 < nb$ (since $nb - na > 1$). Setting $q = m/n$ gives $a < q < b$ with $q \in \mathbb{Q}$. □

**Theorem 3.12** (Nested Intervals Property)**.** *Let $I_n = [a_n, b_n]$ be a sequence of closed bounded intervals with $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$. Then $\bigcap_{n=1}^\infty I_n \neq \emptyset$.*

*Proof.* The nesting condition gives $a_1 \leq a_2 \leq \cdots$ and $b_1 \geq b_2 \geq \cdots$ with $a_n \leq b_n$ for all $n$. In particular, $a_n \leq b_1$ for all $n$, so $\{a_n : n \in \mathbb{N}\}$ is bounded above. By the completeness axiom, $\alpha = \sup\{a_n : n \in \mathbb{N}\}$ exists.

For every $n$, we have $a_n \leq \alpha$ (since $\alpha$ is an upper bound) and $\alpha \leq b_n$ (since $b_n$ is an upper bound for $\{a_k\}$: for $k \leq n$, $a_k \leq a_n \leq b_n$; for $k > n$, $a_k \leq b_k \leq b_n$). Therefore $\alpha \in [a_n, b_n]$ for every $n$, giving $\alpha \in \bigcap_{n=1}^\infty I_n$. □

**Closure and boundedness are both essential**

The nested intervals conclusion fails for open intervals: $\bigcap_{n=1}^\infty (0, 1/n) = \emptyset$. It also fails for unbounded closed sets: $\bigcap_{n=1}^\infty [n, \infty) = \emptyset$. Both closure *and* boundedness are required.

**Example 3.13.** The bisection method for root-finding constructs nested intervals $[a_n, b_n]$ with $f(a_n) < 0 < f(b_n)$ and $b_n - a_n = (b_0 - a_0)/2^n$. The nested intervals property guarantees a point in $\bigcap_n [a_n, b_n]$, and continuity of $f$ forces $f$ to vanish there.

With the real number system and its completeness properties established, we now turn to the central objects of analysis: sequences and their convergence behavior.

## 4 Sequences

**Definition 4.1** (Convergence)**.** A sequence $(a_n)_{n=1}^{\infty}$ in $\mathbb{R}$ *converges* to $L \in \mathbb{R}$, written $\lim_{n\to\infty} a_n = L$ or $a_n \to L$, if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for all $n \geq N$. A sequence that does not converge is *divergent*.

**Example 4.2.** The sequence $a_n = 1/n$ converges to 0. Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $N > 1/\varepsilon$ (Archimedean property). For $n \geq N$: $|a_n - 0| = 1/n \leq 1/N < \varepsilon$.

**Theorem 4.3** (Uniqueness of Limits)**.** *If $a_n \to L$ and $a_n \to M$, then $L = M$.*

*Proof.* Let $\varepsilon > 0$. There exist $N_1, N_2 \in \mathbb{N}$ such that $|a_n - L| < \varepsilon/2$ for $n \geq N_1$ and $|a_n - M| < \varepsilon/2$ for $n \geq N_2$. For $n \geq \max(N_1, N_2)$:

$$|L - M| \leq |L - a_n| + |a_n - M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $|L - M| < \varepsilon$ for every $\varepsilon > 0$, we conclude $L = M$. □

**Theorem 4.4** (Algebraic Limit Theorems)**.** *Suppose $a_n \to a$ and $b_n \to b$. Then:*

*(i)* $a_n + b_n \to a + b$.
*(ii)* $a_n \cdot b_n \to a \cdot b$.
*(iii) If $b \neq 0$, then $a_n/b_n \to a/b$ (for $n$ sufficiently large that $b_n \neq 0$).*

*Proof.* We prove (ii); parts (i) and (iii) follow by similar arguments.

Since $(a_n)$ converges, it is bounded: there exists $M > 0$ with $|a_n| \leq M$ for all $n$. Let $\varepsilon > 0$. Choose $N_1$ so that $|a_n - a| < \varepsilon/(2(|b| + 1))$ for $n \geq N_1$, and $N_2$ so that $|b_n - b| < \varepsilon/(2M)$ for $n \geq N_2$. For $n \geq \max(N_1, N_2)$:

$$\begin{aligned} |a_n b_n - ab| &= |a_n b_n - a_n b + a_n b - ab| \\ &\leq |a_n||b_n - b| + |b||a_n - a| \\ &\leq M \cdot \frac{\varepsilon}{2M} + |b| \cdot \frac{\varepsilon}{2(|b| + 1)} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

□

**Example 4.5.** Consider the sequence $a_n = (3n^2 + 1)/(n^2 + 5n)$. Dividing numerator and denominator by $n^2$:

$$a_n = \frac{3 + 1/n^2}{1 + 5/n}.$$

Since $1/n^2 \to 0$ and $5/n \to 0$, the algebraic limit theorems give $a_n \to 3/1 = 3$.

**Monotone Convergence Theorem**

**Theorem 4.6** (Monotone Convergence Theorem for Sequences)**.** *Every bounded monotone sequence converges. Specifically:*

*(i) If $(a_n)$ is increasing and bounded above, then $a_n \to \sup\{a_n : n \in \mathbb{N}\}$.*
*(ii) If $(a_n)$ is decreasing and bounded below, then $a_n \to \inf\{a_n : n \in \mathbb{N}\}$.*

*Proof.* We prove (i). Let $\alpha = \sup\{a_n : n \in \mathbb{N}\}$, which exists by the completeness axiom. Let $\varepsilon > 0$. Since $\alpha - \varepsilon$ is not an upper bound for $\{a_n\}$, there exists $N$ with $a_N > \alpha - \varepsilon$. Since $(a_n)$ is increasing, for $n \geq N$:

$$\alpha - \varepsilon < a_N \leq a_n \leq \alpha,$$

so $|a_n - \alpha| < \varepsilon$. Part (ii) follows by applying (i) to $(-a_n)$. □

**Monotone convergence in policy iteration**

Policy iteration generates a sequence of value functions $V^{\pi_0}, V^{\pi_1}, V^{\pi_2}, \ldots$ that is monotonically non-decreasing (each policy improvement step satisfies $V^{\pi_{k+1}}(s) \geq V^{\pi_k}(s)$ for all $s$) and bounded above by $V^*(s) \leq R_{\max}/(1-\gamma)$. The monotone convergence theorem guarantees convergence of this sequence.

**Theorem 4.7** (Bolzano–Weierstrass)**.** *Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.*

*Proof.* Let $(a_n)$ be bounded. We construct a monotone subsequence, which converges by Theorem 4.6.

Call an index $n$ a *peak* if $a_n \geq a_m$ for all $m > n$.

**Case 1:** There are infinitely many peaks $n_1 < n_2 < n_3 < \cdots$. Then $a_{n_1} \geq a_{n_2} \geq a_{n_3} \geq \cdots$ is a decreasing subsequence. It is bounded (since $(a_n)$ is bounded), hence convergent by Theorem 4.6.

**Case 2:** There are only finitely many peaks. Let $N$ be larger than all peak indices. Then no index $n \geq N$ is a peak, meaning for every $n \geq N$ there exists $m > n$ with $a_m > a_n$. Starting with $n_1 = N$, inductively choose $n_{k+1} > n_k$ with $a_{n_{k+1}} > a_{n_k}$. This gives a strictly increasing subsequence. It is bounded above (since $(a_n)$ is bounded), hence convergent by Theorem 4.6. □

**Example 4.8.** The sequence $a_n = (-1)^n(1 + 1/n)$ is bounded but divergent. However, the subsequence $a_{2k} = 1 + 1/(2k) \to 1$ converges, as guaranteed by Bolzano–Weierstrass.

**Definition 4.9** (Cauchy Sequence)**.** A sequence $(a_n)$ in $\mathbb{R}$ is *Cauchy* if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - a_m| < \varepsilon$ for all $n, m \geq N$.

**Theorem 4.10** (Completeness of $\mathbb{R}$: Cauchy Criterion)**.** *A sequence in $\mathbb{R}$ converges if and only if it is Cauchy.*

*Proof.* ($\Rightarrow$) Suppose $a_n \to L$. Given $\varepsilon > 0$, choose $N$ so that $|a_n - L| < \varepsilon/2$ for $n \geq N$. For $n, m \geq N$:

$$|a_n - a_m| \leq |a_n - L| + |L - a_m| < \varepsilon.$$

($\Leftarrow$) Suppose $(a_n)$ is Cauchy. First, $(a_n)$ is bounded: choose $N$ so that $|a_n - a_m| < 1$ for $n, m \geq N$; then $|a_n| \leq \max(|a_1|, \ldots, |a_{N-1}|, |a_N| + 1)$ for all $n$.

By Bolzano–Weierstrass (Theorem 4.7), $(a_n)$ has a convergent subsequence $a_{n_k} \to L$. We show $a_n \to L$. Given $\varepsilon > 0$, choose $N_1$ so that $|a_n - a_m| < \varepsilon/2$ for $n, m \geq N_1$, and choose $K$ so that $|a_{n_k} - L| < \varepsilon/2$ for $k \geq K$. Pick $k \geq K$ with $n_k \geq N_1$. Then for $n \geq N_1$:

$$|a_n - L| \leq |a_n - a_{n_k}| + |a_{n_k} - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$ □

**Cauchy sequences in iterative algorithms**

The Cauchy criterion is indispensable in RL and optimization because it certifies convergence *without knowing the limit*. In practice, we often cannot compute $V^*$ or $\theta^*$ directly. Instead, we show successive iterates $\|V_{n+1} - V_n\|$ shrink geometrically (e.g., by a factor

$\gamma$), verifying the Cauchy condition, and conclude that the limit exists. This is precisely the logic behind contraction-mapping proofs of value iteration convergence.

**Definition 4.11** (lim sup and lim inf)**.** For a bounded sequence $(a_n)$, define:

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty}\Big(\sup_{k\geq n} a_k\Big), \qquad \liminf_{n\to\infty} a_n = \lim_{n\to\infty}\Big(\inf_{k\geq n} a_k\Big).$$

The sequence $\sup_{k\geq n} a_k$ is decreasing and bounded below (by $\inf a_k$), and $\inf_{k\geq n} a_k$ is increasing and bounded above (by $\sup a_k$), so both limits exist by Theorem 4.6.

**Example 4.12.** For $a_n = (-1)^n + 1/n$, the odd-indexed terms approach $-1$ from above and the even-indexed terms approach 1 from above. We have $\sup_{k\geq n} a_k \to 1$ and $\inf_{k\geq n} a_k \to -1$, so $\limsup a_n = 1$ and $\liminf a_n = -1$. Since these differ, the sequence diverges.

**Proposition 4.13** (Properties of lim sup and lim inf)**.** *Let $(a_n)$ be a bounded sequence. Then:*

*(i)* $\liminf_{n\to\infty} a_n \leq \limsup_{n\to\infty} a_n$.

*(ii)* $(a_n)$ *converges if and only if* $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n$*, and in that case the common value equals* $\lim a_n$.

*(iii)* $\limsup_{n\to\infty} a_n = L$ *if and only if: for every* $\varepsilon > 0$*, (a)* $a_n < L + \varepsilon$ *for all sufficiently large* $n$*, and (b)* $a_n > L - \varepsilon$ *for infinitely many* $n$.

*Proof.* (i) For each $n$, $\inf_{k\geq n} a_k \leq \sup_{k\geq n} a_k$. Taking limits preserves the inequality.

(ii) ($\Rightarrow$) Suppose $a_n \to L$. Given $\varepsilon > 0$, for large $n$ we have $L - \varepsilon < a_k < L + \varepsilon$ for all $k \geq n$, so $L - \varepsilon \leq \inf_{k\geq n} a_k \leq \sup_{k\geq n} a_k \leq L + \varepsilon$. Taking $n \to \infty$ gives $L - \varepsilon \leq \liminf a_n \leq \limsup a_n \leq L + \varepsilon$ for all $\varepsilon > 0$, so both equal $L$.

($\Leftarrow$) Suppose $\liminf a_n = \limsup a_n = L$. For any $\varepsilon > 0$, choose $N$ so that $|\sup_{k\geq n} a_k - L| < \varepsilon$ and $|\inf_{k\geq n} a_k - L| < \varepsilon$ for $n \geq N$. Then for $n \geq N$:

$$L - \varepsilon < \inf_{k\geq n} a_k \leq a_n \leq \sup_{k\geq n} a_k < L + \varepsilon,$$

so $|a_n - L| < \varepsilon$.

(iii) Let $s_n = \sup_{k\geq n} a_k$ and $L = \lim s_n$.

For (a): given $\varepsilon > 0$, choose $N$ with $s_N < L + \varepsilon$. Then $a_n \leq s_N < L + \varepsilon$ for all $n \geq N$.

For (b): given $\varepsilon > 0$, for every $n$ we have $s_n \geq L$ (since $s_n$ decreases to $L$), so there exists $k \geq n$ with $a_k > s_n - \varepsilon \geq L - \varepsilon$. Since $n$ is arbitrary, infinitely many such $k$ exist.

Conversely, suppose (a) and (b) hold for some value $L'$. From (a), $s_n \leq L' + \varepsilon$ for large $n$, so $\limsup a_n \leq L' + \varepsilon$. From (b), for each $n$ there exists $k \geq n$ with $a_k > L' - \varepsilon$, so $s_n \geq L' - \varepsilon$, giving $\limsup a_n \geq L' - \varepsilon$. Since $\varepsilon$ is arbitrary, $\limsup a_n = L'$. □

**lim sup is not sup**

A common mistake is to confuse $\limsup_{n\to\infty} a_n$ with $\sup_n a_n$. The supremum $\sup_n a_n$ is the largest value attained (or approached) by the entire sequence, while $\limsup a_n$ captures the largest *cluster point*—the largest value that subsequences approach. For instance, the sequence $a_1 = 100, a_n = (-1)^n$ for $n \geq 2$ has $\sup a_n = 100$ but $\limsup a_n = 1$.

With the theory of sequences complete—including criteria for convergence, extraction of convergent subsequences, and the lim sup and lim inf tools—we are ready to study infinite series.

## 5 Series

**Definition 5.1** (Convergence of Series and Absolute Convergence)**.** Let $(a_n)_{n=1}^{\infty}$ be a sequence. The *series* $\sum_{n=1}^{\infty} a_n$ *converges* if the sequence of partial sums $S_N = \sum_{n=1}^{N} a_n$ converges. The series *converges absolutely* if $\sum_{n=1}^{\infty} |a_n|$ converges.

**Example 5.2.** The geometric series $\sum_{n=0}^{\infty} r^n$ converges for $|r| < 1$. Indeed, the partial sum $S_N = (1-r^{N+1})/(1-r)$ satisfies $S_N \to 1/(1-r)$, since $|r|^{N+1} \to 0$ by the Archimedean property and algebraic limit theorems.

**Theorem 5.3** (Comparison Test)**.** *If $0 \le a_n \le b_n$ for all $n$ and $\sum b_n$ converges, then $\sum a_n$ converges.*

*Proof.* The partial sums $S_N = \sum_{n=1}^{N} a_n$ form an increasing sequence (since $a_n \ge 0$) bounded above by $\sum_{n=1}^{\infty} b_n < \infty$. By the monotone convergence theorem (Theorem 4.6), $(S_N)$ converges. □

**Theorem 5.4** (Absolute Convergence Implies Convergence)**.** *If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.*

*Proof.* Absolute convergence means the partial sums of $|a_n|$ form a Cauchy sequence. For $m > n$:

$$\Big|\sum_{k=n+1}^{m} a_k\Big| \le \sum_{k=n+1}^{m} |a_k|,$$

so the partial sums of $a_n$ are also Cauchy, hence convergent by Theorem 4.10. □

**Example 5.5.** The series $\sum_{n=1}^{\infty}(-1)^n/n^2$ converges absolutely, since $\sum 1/n^2$ converges (this can be shown by comparison with $\sum 1/(n(n-1))$, which telescopes to 1). By Theorem 5.4, the original alternating series converges.

**Ratio and Root Tests**

**Theorem 5.6** (Ratio Test)**.** *Let $(a_n)$ be a sequence with $a_n \neq 0$ for all $n$.*
*(i) If $\limsup_{n\to\infty} |a_{n+1}/a_n| < 1$, then $\sum a_n$ converges absolutely.*
*(ii) If $\liminf_{n\to\infty} |a_{n+1}/a_n| > 1$, then $\sum a_n$ diverges.*

**Theorem 5.7** (Root Test)**.** *Let $\alpha = \limsup_{n\to\infty} |a_n|^{1/n}$.*
*(i) If $\alpha < 1$, then $\sum a_n$ converges absolutely.*
*(ii) If $\alpha > 1$, then $\sum a_n$ diverges.*

*Proof of the Ratio Test.* Suppose $\limsup |a_{n+1}/a_n| = L < 1$. Choose $r$ with $L < r < 1$. By Proposition 4.13(iii), there exists $N$ such that $|a_{n+1}/a_n| < r$ for all $n \ge N$. Then $|a_n| \le |a_N| r^{n-N}$ for $n \ge N$, and $\sum |a_N| r^{n-N}$ converges (geometric series with ratio $r < 1$). By the comparison test, $\sum |a_n|$ converges.

If $\liminf |a_{n+1}/a_n| > 1$, then $|a_{n+1}| > |a_n|$ for all sufficiently large $n$, so $a_n \not\to 0$ and the series diverges (since convergence requires $a_n \to 0$). □

*Proof of the Root Test.* If $\alpha < 1$, choose $r$ with $\alpha < r < 1$. By Proposition 4.13(iii), $|a_n|^{1/n} < r$ for all sufficiently large $n$, so $|a_n| < r^n$. The comparison test with the geometric series $\sum r^n$ gives absolute convergence.

If $\alpha > 1$, then $|a_n|^{1/n} > 1$ for infinitely many $n$ (Proposition 4.13(iii)), so $|a_n| > 1$ infinitely often, meaning $a_n \not\to 0$ and the series diverges. □

**Example 5.8.** Consider the series $\sum_{n=0}^{\infty} x^n/n!$ (the exponential series). The ratio test gives $|a_{n+1}/a_n| = |x|/(n+1) \to 0 < 1$, so the series converges absolutely for every $x \in \mathbb{R}$. This defines $e^x$ as a convergent power series.

**Series convergence in function approximation**

In supervised learning, kernel methods express predictions as $f(x) = \sum_{n=1}^{\infty} \alpha_n \phi_n(x)$ for basis functions $\phi_n$. Convergence of this series (and control of the approximation error from truncating to finitely many terms) is established via the comparison, ratio, or root tests applied to the coefficient sequence $(\alpha_n)$.

*Remark* 5.9 (Conditional Convergence). A series that converges but does not converge absolutely is called *conditionally convergent*. The alternating harmonic series $\sum(-1)^{n+1}/n$ is the classic example. By the Riemann rearrangement theorem, any conditionally convergent series can be rearranged to converge to any prescribed real value, or to diverge. This pathology motivates the focus on absolute convergence in applications. In RL and numerical analysis, one almost always works with absolutely convergent series to ensure robustness under reordering of summation.

## 6 Exercises

**Exercise 6.1** (Supremum computation). $(*)$
Let $S = \{1 - 1/n : n \in \mathbb{N}\}$. Prove that $\sup S = 1$ directly from the definition of supremum (Definition 3.4).

**Exercise 6.2** (Convergence from the definition). $(*)$
Prove directly from Definition 4.1 that the sequence $a_n = (2n+1)/(n+3)$ converges to 2.

**Exercise 6.3** (Monotone convergence application). $(**)$
Define the sequence $a_1 = 1$, $a_{n+1} = \sqrt{2 + a_n}$. Prove that $(a_n)$ is increasing and bounded above, and find its limit.

*Hint:* Show by induction that $a_n < 2$ for all $n$. For the limit, if $a_n \to L$ then $L = \sqrt{2+L}$.

**Exercise 6.4** (Cauchy criterion in RL). $(**)$
Let $T\colon \mathbb{R} \to \mathbb{R}$ be a contraction mapping with factor $\gamma \in (0,1)$: $|Tx - Ty| \le \gamma|x - y|$ for all $x, y \in \mathbb{R}$. Define $x_0 \in \mathbb{R}$ arbitrarily and $x_{n+1} = Tx_n$. Prove that $(x_n)$ is a Cauchy sequence, and conclude that it converges.

*Hint:* Show $|x_{n+1} - x_n| \le \gamma^n |x_1 - x_0|$, then use the geometric series formula for $|x_n - x_m|$.

**Exercise 6.5** (Series convergence tests). $(*)$
Determine whether each series converges or diverges, and state which test you use:

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{3^n}{n!}$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{n}{2^n}$

(c) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$

**Exercise 6.6** (Limsup and subsequential limits). $(***)$
Let $(a_n)$ be a bounded sequence. Prove that there exists a subsequence $(a_{n_k})$ with $a_{n_k} \to \limsup_{n\to\infty} a_n$.

*Hint:* Use the characterization in Proposition 4.13(iii) to inductively select $n_k$ with $|a_{n_k} - L| < 1/k$ where $L = \limsup a_n$.

## 7 Summary

- The real numbers $\mathbb{R}$ form an ordered field satisfying the *completeness axiom*: every nonempty subset bounded above has a supremum. This axiom is what distinguishes $\mathbb{R}$ from $\mathbb{Q}$.
- The completeness axiom implies the *greatest lower bound property*: every nonempty subset bounded below has an infimum (Corollary 3.8).
- The *Archimedean property* ($\mathbb{N}$ is unbounded) and the *density of rationals* are consequences of completeness.
- A sequence $(a_n)$ converges to $L$ if the terms eventually stay within any prescribed distance $\varepsilon$ of $L$; limits are unique and respect algebraic operations.
- The *monotone convergence theorem* states that every bounded monotone sequence converges; this directly underpins policy iteration convergence in RL.
- The *Bolzano–Weierstrass theorem* guarantees that every bounded sequence has a convergent subsequence, a cornerstone for compactness arguments.
- A sequence converges if and only if it is *Cauchy* (completeness of $\mathbb{R}$). The Cauchy criterion enables convergence proofs for iterative algorithms without knowing the limit.
- The $\limsup$ and $\liminf$ of a bounded sequence always exist; the sequence converges exactly when these two values coincide.
- For series, the *comparison test*, *ratio test*, and *root test* provide practical tools for establishing absolute convergence. Absolute convergence implies convergence and guards against rearrangement pathologies.

## 8 Further Reading

- **Lesson 2** (Continuity, Differentiation, and Integration) builds on the convergence theory developed here. The intermediate value theorem, mean value theorem, and Riemann integration all rely on completeness and the Bolzano–Weierstrass theorem.
- **Lesson 3** (Topology of the Real Line and Metric Spaces) generalizes convergence to abstract metric spaces. Compactness in metric spaces is the natural extension of the Bolzano–Weierstrass property.
- **Lesson 4** (Sequences and Series of Functions) treats pointwise and uniform convergence, power series, and the Weierstrass M-test—a direct generalization of the comparison test to function series.
- For a deeper treatment of the construction of $\mathbb{R}$ from $\mathbb{Q}$ via Dedekind cuts or Cauchy sequences, see Rudin [1976, Chapter 1] or Tao [2016, Chapters 1–5].
- Students proceeding to Phase 01 (measure theory) will find that the monotone convergence theorem and Cauchy completeness generalize to Lebesgue integration, where the monotone convergence theorem for integrals and the completeness of $L^p$ spaces play analogous roles.
- Pugh [2015] provides an enriched treatment with many challenging exercises and a geometric perspective on real analysis that complements the more algebraic approach of Rudin [1976].

## References

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